Odd-zeta linear forms and their arithmetic cost

A derivational note accompanying the Lean development

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Abstract

A positive moment determinant isolates any fixed odd zeta value without a weight projector. We derive its nonvanishing, construct explicit integer coefficients, and account for denominator clearing. For a fixed moment shift, the raw determinant has logarithmic decay rate $\log 4$ on the square of its size. An elementary binomial divisibility bound shows that the uniform least-common-multiple normalization instead grows at every integer weight $s > 1$. Exact common-factor cancellation is therefore essential for this family. The accompanying Lean file verifies the integer construction and its conditional irrationality criterion. A Jacobi comparison gives explicit two-sided estimates for the approximation error Δ/A, uniformly in all nonnegative shifts. The remaining condition becomes a precise bound on the primitive target coefficient. No unconditional irrationality theorem for all odd zeta values is established.

1 The quantity an irrationality argument must make small

Write a rational linear form as $L = A\alpha + B$, and choose an integer $D > 0$ such that $a = DA$ and $b = DB$ are integers. If $(a, b) \neq (0, 0)$, set $g = \gcd(|a|, |b|)$ and

$$\ell = \frac{D}{g}L = \frac{a}{g}\alpha + \frac{b}{g}. \tag{1}$$

Under the hypothesis $\alpha = p/q \in \mathbb{Q}$, every nonzero ℓ satisfies $|q\ell| \geq 1$. Thus arbitrarily late nonzero forms tending to zero imply irrationality. The exact arithmetic budget is

$$-\log|\ell| = -\log|L| - \log D + \log g. \tag{2}$$

An analytic estimate on L alone does not control this budget. The common divisor in (1) must be proved, rather than inferred from a rational or geometric change of coordinates.

2 A pure-target construction at every integer weight

Fix integers $s > 1$, $a \geq 0$, and $N \geq 1$. Define

$$H_k^{(s)} = \sum_{j=1}^{k} j^{-s}, \qquad \mu_k = \zeta(s) - H_k^{(s)}, \qquad \Delta_{N,a}^{(s)} = \det[\mu_{a+i+j}]_{0\leq i,j<N}. \tag{3}$$

The case requested in the chat is $s = 2r + 1$ with $r \geq 1$.

Theorem 1. *For these parameters there are explicit $A_{N,a}^{(s)}, B_{N,a}^{(s)} \in \mathbb{Q}$ such that*

$$0 < \Delta_{N,a}^{(s)} = A_{N,a}^{(s)}\zeta(s) + B_{N,a}^{(s)}. \tag{4}$$

In particular, this construction isolates one target and is nonzero.

Proof. Substitute $x = e^{-t}$ in the elementary gamma integral to obtain

$$\frac{1}{(j+1)^s} = \frac{1}{(s-1)!}\int_0^1 x^j(-\log x)^{s-1}\,\mathrm{d}x.$$

Summing nonnegative integrands gives

$$\mu_k = \int_0^1 x^k w_s(x)\,\mathrm{d}x, \qquad w_s(x) = \frac{(-\log x)^{s-1}}{(s-1)!(1-x)}. \tag{5}$$

This measure is finite: its total mass is $\zeta(s) < \infty$. Expanding the determinant and symmetrizing its product integrals gives

$$\Delta_{N,a}^{(s)} = \frac{1}{N!}\int_{(0,1)^N}\prod_{i<j}(x_i - x_j)^2\prod_{i=1}^N x_i^a w_s(x_i)\,\mathrm{d}x_i > 0. \tag{6}$$

The integrand is bounded above by the product of the finite measures. It is strictly positive when the variables lie in pairwise disjoint closed interior intervals, proving strict positivity of the integral.

For the affine identity, put

$$P(x) = \det[x - H_{a+i+j}^{(s)}]_{0\le i,j<N}.$$

Subtract row zero from every other row. Only row zero then depends on x, so determinant multilinearity gives $P(x) = (P(1) - P(0))x + P(0)$. Take $A = P(1) - P(0)$ and $B = P(0)$. □

The coefficient of the target has a useful independent expression:

$$A_{N,a}^{(s)} = \det\left[\frac{1}{(a+i+j+1)^s} - \frac{1}{(a+i+j+2)^s}\right]_{0\le i,j<N-1} > 0. \tag{7}$$

Here an empty determinant is 1. For $N = 1$ the coefficient is 1 directly. For $N \ge 2$, replace the monomial basis by

$$1, \quad (1-x), \quad x(1-x), \quad \ldots, \quad x^{N-2}(1-x).$$

This integer basis change has determinant ± 1. Only the first polynomial is nonzero at 1, so the coefficient of $\zeta(s)$ is the lower principal minor. Its entries are

$$\mu_k - 2\mu_{k+1} + \mu_{k+2} = (k+1)^{-s} - (k+2)^{-s}.$$

The same positive Gram argument proves the inequality.

3 Explicit error estimates, uniform even for growing shifts

Put $s = 2r+1$, $r \ge 1$, and write $(z)_k = z(z+1)\cdots(z+k-1)$. The following bounds concern the actual coefficients in (4).

Theorem 2. *For $N > r$ and every integer $a \ge 0$,*

$$\frac{(2r-1)!}{(N)_{2r}(N+a)_{2r}} \le E_{N,a} := \frac{\Delta_{N,a}^{(2r+1)}}{A_{N,a}^{(2r+1)}} \le \frac{(2r-1)!}{(N-r)_{2r}(N-r+a)_{2r}}. \tag{8}$$

The lower bound holds for all $N \ge 1$. For every sequence $a_N \ge 0$,

$$N^{2r}(N+a_N)^{2r}E_{N,a_N} \longrightarrow (2r-1)!. \tag{9}$$

Proof. The positive-functional method of [1] suggests using the approximation error itself. The basis change preceding (7) and the Schur complement of its lower block give the exact minimum

$$E_{N,a} = \min_{\deg P<N,\ P(1)=1} \int_0^1 P(x)^2 \rho(x)\,\mathrm{d}x, \quad \rho(x) = \frac{x^a(-\log x)^{2r}}{(2r)!(1-x)}. \tag{10}$$

Indeed, every admissible P has the unique form $1+(1-x)Q$, $\deg Q < N-1$; completing the square in the positive Gram matrix gives its determinant divided by the lower-block determinant A.

For nonnegative integers m, b, the corresponding minimum for $x^b(1-x)^m$ is

$$\lambda_N^{(m,b)}(1) = \frac{m!(m+1)!}{(N)_{m+1}(N+b)_{m+1}}. \tag{11}$$

To check the constant, use the shifted Jacobi polynomials

$$J_j(x) = \frac{(-1)^j}{j!x^b(1-x)^m} \frac{\mathrm{d}^j}{\mathrm{d}x^j}\left[x^{b+j}(1-x)^{m+j}\right].$$

Integrating by parts j times gives orthogonality and, using the beta integral, their squared norms

$$h_j = \frac{(j+m)!(j+b)!}{(2j+m+b+1)j!(j+m+b)!}, \qquad J_j(1) = \binom{j+m}{j}.$$

All boundary terms vanish by the endpoint zero orders. Cauchy–Schwarz in this basis shows that the minimum is $1/\sum_{j<N} J_j(1)^2/h_j$, with equality attained by coefficients proportional to $J_j(1)/h_j$. The sum telescopes to $(N)_{m+1}(N+b)_{m+1}/(m!(m+1)!)$; subtracting its values at $N = j+1$ and $N = j$ verifies each summand. This proves (11).

The inequalities $1-x \le -\log x \le (1-x)/x$ imply, for $q(x) = x^a(1-x)^{2r-1}/(2r)!$, that $q \le \rho$ and $x^{2r}\rho \le q$. The first gives the lower bound in (8) by (11). For the upper bound restrict (10) to $P = x^r Q$, $\deg Q < N-r$, $Q(1) = 1$; then $\int P^2\rho \le \int Q^2 q$. Apply (11) again.

Finally $N^k \le (N)_k \le (N+k-1)^k$ shows that the scaled expression in (9) lies between

$$(2r-1)!\left(\frac{N}{N+2r-1}\right)^{4r} \quad \text{and} \quad (2r-1)!\left(\frac{N}{N-r}\right)^{4r}.$$

These bounds are independent of $a \ge 0$ and both tend to $(2r-1)!$. □

4 Explicit denominator clearing, including its cost

Let $M = a+2N-2$, $d_M = \mathrm{lcm}(1,\dots,M)$, with $d_0 = 1$, and put $Q = d_M^s$. For every $m \le M$,

$$QH_m^{(s)} = \sum_{j=1}^{m} (d_M/j)^s \in \mathbb{Z}. \tag{12}$$

Define the integer matrix and two integers

$$Z_{ij} = -\sum_{j'=1}^{a+i+j} (d_M/j')^s, \tag{13}$$

$$U = \det[Q+Z_{ij}] - \det[Z_{ij}], \qquad V = \det[Z_{ij}]. \tag{14}$$

Applying the affine determinant identity after scaling gives the exact relation

$$\boxed{Q^N \Delta_{N,a}^{(s)} = U\zeta(s) + V, \qquad U, V \in \mathbb{Z}.} \tag{15}$$

Consequently the primitive form is $(Q^N/g)\Delta^{(s)}_{N,a}$, where $g = \gcd(|U|, |V|)$. Writing $u = U/g > 0$, it is also exactly $uE_{N,a}$. Thus (9) proves the sharp equivalent condition

$$\ell_{N,a_N} \to 0 \quad \Longleftrightarrow \quad \frac{u_{N,a_N}}{N^{2r}(N + a_N)^{2r}} \to 0. \tag{16}$$

This primitive-height estimate is not established here. These definitions, including (12) and (15), are proved in the accompanying Lean module for every natural s. The positivity statement uses $s > 1$ and is proved above in ordinary mathematics.

For $M \geq 1$, its exact prime cost is $v_p(Q^N) = sN\lfloor \log_p M \rfloor$. The scale Q^N is safe, but it is expensive. To quantify this without the prime number theorem, observe that

$$\binom{M}{k} \mid d_M \quad (0 \leq k \leq M), \qquad 2^M \leq (M + 1)d_M. \tag{17}$$

The case $M = 0$ is immediate. For $M \geq 1$, Legendre's formula gives, for every prime p,

$$v_p\binom{M}{k} = \sum_{\nu \geq 1} \left(\left\lfloor \frac{M}{p^\nu} \right\rfloor - \left\lfloor \frac{k}{p^\nu} \right\rfloor - \left\lfloor \frac{M - k}{p^\nu} \right\rfloor \right) \leq \lfloor \log_p M \rfloor = v_p(d_M).$$

Each summand is 0 or 1 and is zero for $p^\nu > M$. Summing the $M + 1$ binomial coefficients proves the second inequality. It follows that

$$\log(Q^N) \geq sN\big(M \log 2 - \log(M + 1)\big). \tag{18}$$

There is a proved arithmetic improvement to this uniform scale. In row i all harmonic indices are at most $M_i = a + i + N - 1$, so $d^s_{M_i}$ clears that entire row. Set

$$D_{\mathrm{row}} = \prod_{i=0}^{N-1} d^s_{M_i}, \qquad G_{\mathrm{row}} = \prod_{i=0}^{N-1} (d_M/d_{M_i})^s. \tag{19}$$

Every quotient is integral and $G_{\mathrm{row}}D_{\mathrm{row}} = Q^N$. Define $W(y)_{ij} = d^s_{M_i}y - \sum_{k=1}^{a+i+j}(d_{M_i}/k)^s$, and set $U' = \det W(1) - \det W(0)$, $V' = \det W(0)$. Factoring each row gives

$$D_{\mathrm{row}}\Delta = U'\zeta(s) + V', \quad U = G_{\mathrm{row}}U', \quad V = G_{\mathrm{row}}V', \quad g = G_{\mathrm{row}} \gcd(|U'|, |V'|). \tag{20}$$

Thus G_{row} is a constructed common divisor. Its remaining cost is explicit, including every prime:

$$v_p(D_{\mathrm{row}}) = s \sum_{i=0}^{N-1} \lfloor \log_p M_i \rfloor, \qquad \log D_{\mathrm{row}} \geq s \sum_{i=0}^{N-1} \big(M_i \log 2 - \log(M_i + 1)\big),$$

where a zero index contributes zero to the prime sum. Lean verifies the integer coefficient and gcd factorizations and both finite cost formulas. The final primitive pair is unchanged; (16) still requires control of the remaining gcd.

5 Gcd normalization as a derived lattice operation

Keep the integer row (U, V) from (14), and let $g = \gcd(|U|, |V|) > 0$. Place it in cochain degrees zero and one:

$$C^\bullet = [\mathbb{Z}^2 \xrightarrow{[U\ V]} \mathbb{Z}].$$

Write $u = U/g$, $v = V/g$, and choose integers b, c with $bu + cv = 1$. The matrix

$$S = \begin{pmatrix} b & -v \\ c & u \end{pmatrix}, \qquad \det S = 1, \qquad [U\ V]S = [g\ 0]$$

gives $H^0(C) = \mathbb{Z}$ and $H^1(C) = \mathbb{Z}/g\mathbb{Z}$. Equivalently, the image ideal and the zeroth Fitting ideal of its cokernel are $(U, V) = (g)$. This is an exact integral presentation, retaining the discrete coefficient data.

For a nonzero integer d, the décalage operation of [4, Section 6] is computed on a torsion-free complex by

$$(\eta_d C)^i = \{x \in d^i C^i : \partial x \in d^{i+1} C^{i+1}\}.$$

Its degree-zero term is all of $\mathbb{Z}^2$ exactly when $d \mid U$ and $d \mid V$; its degree-one term is $d\mathbb{Z}$. Thus, whenever these divisibilities hold, identifying $d\mathbb{Z}$ with $\mathbb{Z}$ using the basis element d gives

$$\eta_d C \;\cong\; \big[\mathbb{Z}^2 \xrightarrow{[U/d\ V/d]} \mathbb{Z}\big]. \tag{21}$$

In particular, η_g implements the primitive normalization itself. It changes the complex: its degree-one cohomology is zero, whereas that of C is $\mathbb{Z}/g\mathbb{Z}$. Evaluating the divided row on $(\zeta(s), 1)$ gives exactly the normalized real form in (1).

For each prime p, this bounded free complex has derived completion $\widehat{C}_p = [\mathbb{Z}_p^2 \to \mathbb{Z}_p]$. Its torsion length is

$$e_p = \operatorname{length}_{\mathbb{Z}_p} H^1(\widehat{C}_p) = v_p(g) = \min\{v_p(U), v_p(V)\}, \qquad v_p(0) = +\infty. \tag{22}$$

The valuation pair is precise tropical data for the coefficient row. Exactly e_p successive common-factor divisions by p are available. The same Smith calculation works for a complex of completed integral coefficient sheaves on a pro-étale site. This identifies its cohomology sheaves; global cohomology still depends on the space. Identifying this coefficient complex with the Apéry family's integral regulator complex is an additional comparison. An explicit cellular model is an interval with two vertex stalks and one edge stalk, all $\mathbb{Z}$, and incidence restriction maps $-U, V$. Its cellular coboundary is exactly $[U\ V]$. Relating this coefficient system to a geometric tropicalization requires an integral comparison.

Combining all local lengths recovers the entire arithmetic saving:

$$\log g = \sum_p e_p \log p, \qquad -\log|\ell| = -\log\Delta - \log(Q^N) + \sum_p e_p \log p. \tag{23}$$

Thus a successful geometric construction must prove a quantitative lower bound on these particular torsion lengths while preserving the coefficient row and its real evaluation. Décalage computes the division; it does not by itself prove that the saving in (23) is sufficient. The precise transfer through a framed integral comparison is also elementary: if $E \in \mathrm{GL}_2(\mathbb{Z})$ and

$$[U\ V]E = d[A\ B], \qquad \text{then} \qquad \gcd(|U|, |V|) = |d| \gcd(|A|, |B|).$$

Indeed, E preserves the image ideal of the row, while multiplication by d multiplies its generator by d. Thus an identification transfers a proved content estimate for the source row, with its multiplier retained. This formula, including degenerate rows, is verified in Lean.

6 The exact analytic scale and a normalization obstruction

Theorem 3. *For every fixed integer $s > 1$ and fixed $a \geq 0$,*

$$\lim_{N\to\infty} \frac{\log \Delta_{N,a}^{(s)}}{N^2} = -\log 4. \tag{24}$$

Moreover the uniform integer normalization in (15) satisfies

$$\liminf_{N\to\infty} \frac{\log(Q^N \Delta_{N,a}^{(s)})}{N^2} \geq (s-1)\log 4 > 0. \tag{25}$$

Thus this normalization cannot be the small-form sequence.

Proof. Let $\rho(x) = x^a w_s(x)$. Use $p_0 = 1$ and the monic shifted Chebyshev polynomials

$$p_j(x) = 2^{1-2j} T_j(2x-1) \qquad (j \geq 1).$$

The monic basis change preserves the Gram determinant. Hadamard's inequality, $|T_j| \leq 1$ on $[-1,1]$, and $\int \rho = \mu_a$ yield

$$0 < \Delta_{N,a}^{(s)} \leq \prod_{j=0}^{N-1} \int_0^1 p_j(x)^2 \rho(x)\,\mathrm{d}x \leq \mu_a^N\, 2^{-2(N-1)^2}. \tag{26}$$

This proves the required upper logarithmic limit.

For the lower bound, fix $0 < \varepsilon < 1/2$, let $\lambda = 1 - 2\varepsilon$, and let $c_\varepsilon = \min_{[\varepsilon,1-\varepsilon]} \rho > 0$. Restricting (6) to this interval and making an affine substitution gives

$$\Delta_{N,a}^{(s)} \geq c_\varepsilon^N \lambda^{N^2} J_N, \qquad J_N = \det[(i+j+1)^{-1}]_{i,j<N}. \tag{27}$$

The Cauchy determinant formula, obtained by clearing denominators and factoring its alternating numerator, gives

$$J_N = \prod_{j=0}^{N-1} \frac{(j!)^4}{(2j)!(2j+1)!} = \prod_{j=0}^{N-1} \frac{1}{(2j+1)\binom{2j}{j}^2}.$$

Since $4^j/(2j+1) \leq \binom{2j}{j} \leq 4^j$, the logarithm of this product is $-N(N-1)\log 4 + O(N\log(N+1))$. Hence $\liminf N^{-2}\log\Delta \geq \log\lambda - \log 4$. Let ε decrease to zero to prove (24).

Finally, $M = a + 2N - 2$ and (18) imply $\liminf N^{-2}\log(Q^N) \geq s\log 4$. Adding (24) proves (25). □

This obstruction concerns the uniform scale Q^N. It does not bound the primitive common divisor g. For this fixed-shift family to yield primitive forms tending to zero, the exact requirement is

$$\log g - \log(Q^N) - \log \Delta_{N,a}^{(s)} \longrightarrow +\infty. \tag{28}$$

In particular, (25) forces a saving of at least $(s-1)\log 4$ on the N^2 scale. A rational change of polynomial basis in the same space multiplies both coefficients by $(\det T)^2$; after primitive reduction it gives the same integer pair up to sign. Improving the budget therefore requires additional arithmetic identities or a different polynomial space or measure. A shift $a = a_N$ requires a new asymptotic calculation; theorem (24) assumes a fixed.

7 Checks, source connections, and formalization boundary

The exact rational audit already in the repository clears reduced denominators and removes the entire gcd. At $s = 5$, $N = 20$, $a = 0$, it encloses the primitive form strictly above 1 in absolute value. The displayed rates $-\log|\Delta|/N^2 \approx 1.517981$ and $-\log|\ell|/N^2 \approx -4.772732$ are finite-index diagnostics. They are not used in the proofs above and do not exclude eventual arithmetic improvements. The coefficient-complex audit gives $\log(Q^N)/N^2 \approx 9.053638$ and $\sum_p e_p \log p/N^2 \approx 2.762925$ at this index, recovering the same negative budget via (23). At $s = 9$, $N = 5$, it also detects $e_{11} = 2$ although $11 > M = 8$: common factors can arise beyond the primes in the denominator scale. All these common factors are included in the primitive normalization.

A separate sparse-space audit checks 700 small exponent sets and 60 larger examples across $s = 3, 5, 7, 9$. Every larger example at sizes $4, 6, 8$ has primitive absolute value greater than 1. These finite checks make no asymptotic assertion about other polynomial spaces.

There are genuine published Apéry-family identifications. For the classical V_{12} family, [2, Theorem 1.1, Section 5.4] constructs a regulator period with $LV = -12t$, $v_0 = 2\zeta(3)$, and $v_1 = 10\zeta(3) - 12$. Its operator is

$$L = \theta^3 - t(2\theta+1)(17\theta^2 + 17\theta + 5) + t^2(\theta+1)^3.$$

The integral normal-function cone in [3, Section 5] supplies a geometric comparison target. These results identify the classical period; an all-weight identification of our coefficient presentation and a sufficient bound in (23) are still required.

The formal period-identification step is now complete under its explicit Picard–Fuchs hypotheses. For $F = \sum f_n t^n$ satisfying $LF = ct$, coefficient extraction gives $f_1 = 5f_0 + c$ and

$$(n+2)^3 f_{n+2} = (2n+3)(17(n+1)^2 + 17(n+1) + 5) f_{n+1} - (n+1)^3 f_n.$$

Over any characteristic-zero field these equations determine every coefficient from c and f_0. Consequently

$$\left.\begin{array}{lll} LA = 0, & LB = 12t, & LV = -12t, \\ A(0) = 1, & B(0) = 0, & V(0) = 2z \end{array}\right\} \quad \Longrightarrow \quad V = 2zA - B.$$

Lean verifies this identity for the actual Euler differential operator on formal power series. The published V_{12} calculation supplies $z = \zeta(3)$. A geometric source with the analogous higher-weight equations and integral estimates has not been constructed here.

What the Lean file verifies. `OddHankelArithmetic.lean` defines the harmonic sums, the lcm, the integer matrix, and U, V. It proves (12), (15), the lcm lower bound, the finite cost (18), and its exact prime valuations. It also verifies division by any proved common divisor and the resulting conditional irrationality criterion. `GcdDecalage.lean` constructs the coefficient cochain complex and verifies the image ideal, the décalage source condition, the divided boundary and its cochain comparison map over any commutative coefficient ring, and the Smith and valuation identities. It does not formalize the general derived functor $L\eta$ or pro-étale or geometric tropical comparison theorems. `IntegralFramedComparison.lean` verifies the quotient equivalence for the coefficient cokernel, surjectivity of the primitive row, and the framed content-transfer formula above. `AperyModernIdentification.lean` verifies the displayed operator expansion and the coefficient-identification step under explicit recurrence and initial-value hypotheses. `AperyFormalPeriod.lean` derives those recurrences from the formal Picard–Fuchs equations and proves the resulting full power-series identity. The moment integrals, positivity, and asymptotics (24)–(25) are paper proofs here, not Lean-verified results.
`HankelEndpointEstimates.lean` verifies the logarithm and pointwise weight comparisons used in (8), and the ascending-factorial bounds. The Schur complement, Jacobi integration, and uniform limit are paper proofs. The exact endpoint audit checks 24 cases, including growing shifts.
`RowwiseHankelArithmetic.lean` verifies (20) and the finite prime and logarithmic costs in (19). No unconditional all-odd irrationality conclusion follows.

References

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